

format **short** % long for 15 sig figs, change to short for 4 sig figs

SES

UTS*Area load cases

Segment structure top view cross section between walls:		mm^2	BLANK
10.3.4.b Vertical acceleration $20 \cdot \text{mass} \cdot \text{segments_high}$:	0.00E+00	N	BLANK
Segment compression strength UTS*top_view_area:	0.00E+00	0.00%	BLANK
Segment structure front view cross section between walls:		mm^2	BLANK
10.3.4.b Lateral acceleration $40 \cdot \text{mass} \cdot \text{segments_wide}$:	0.00E+00	N	BLANK
Segment compression strength UTS*top_view_area:	0.00E+00	0.00%	BLANK
Segment structure side view cross section between walls:		mm^2	BLANK
10.3.4.b Longitudinal acceleration $40 \cdot \text{mass} \cdot \text{segments_long}$:	0.00E+00	N	BLANK
Segment compression strength UTS*top_view_area:	0.00E+00	0.00%	BLANK

Module Volume

<https://enepaq.com/wp-content/uploads/2023/09/VTC6-Sony-Murata-Li-ion-Battery-Module-With-Temperature-Sensor-Datasheet-Tesla-Battery-Sponsorship-Formula-SAE-Electric-Formula-Student-Battery-Pack.pdf>

These represent the dimensions of 1 battery module (picture below). Variables are used for other calculations

EneModWidth = $20.3/1000\%(m)$

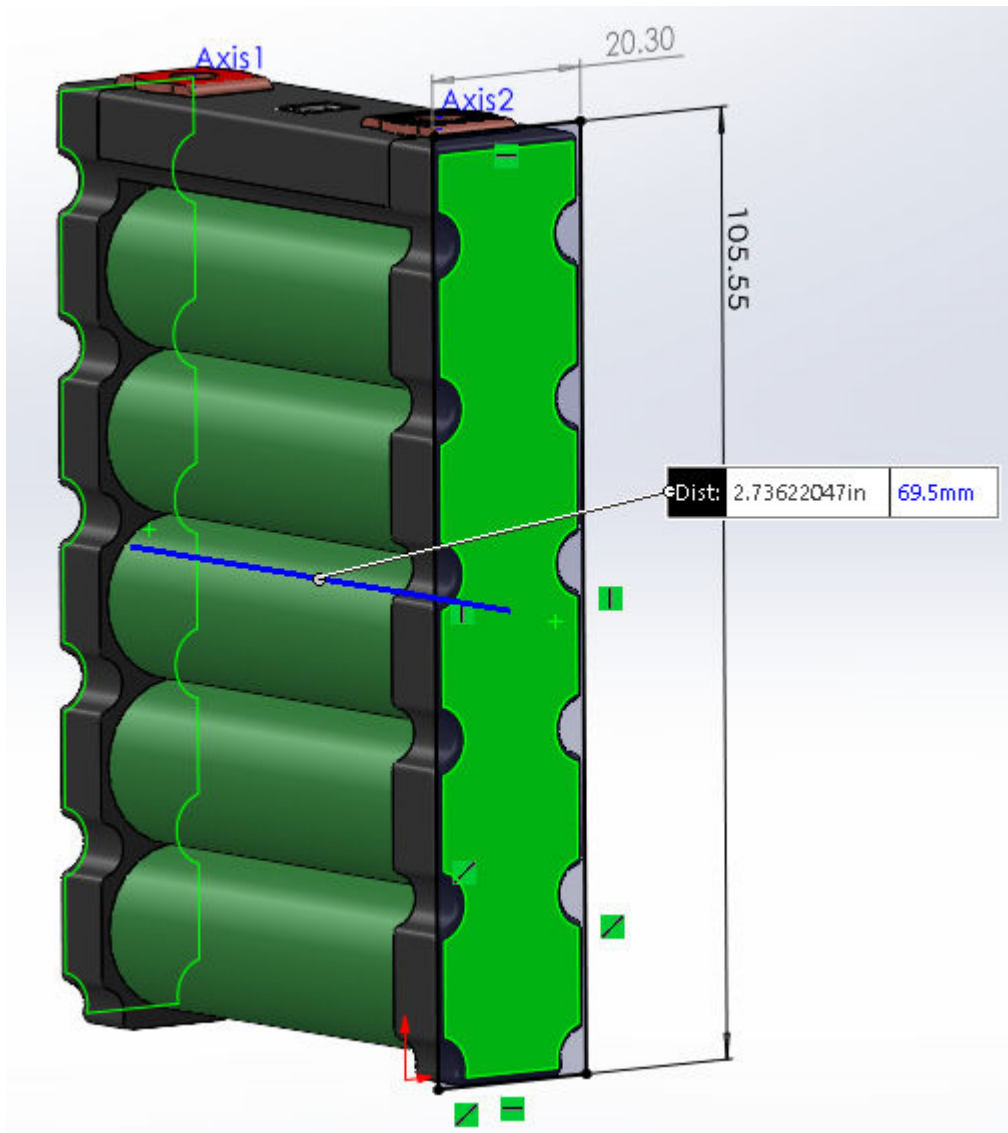
EneModWidth = 0.0203

EneSegHeight = $105.6/1000\%(m)$

EneSegHeight = 0.1056

EneModDepth = $69.5/1000\%(m)$

EneModDepth = 0.0695



ModuleWeight = 278/1000%kg(aka 278g)

ModuleWeight = 0.2780

Segment Dimensions

SegmentCasingHeight = EneSegHeight%m(Segment Height is pretty much the height of one module)

SegmentCasingHeight = 0.1056

EneSegmentDepth = 6*EneModDepth%num moduleSs*moduleDepth*mmtoinches + flange added length - hole location

EneSegmentDepth = 0.4170

EneSegmentWidth = EneModWidth * 4% rough width of segment

EneSegmentWidth = 0.0812

$$\text{garoliteThickness} = (1/16) \cdot 25.4 / 1000 \text{ m}$$

$$\text{garoliteThickness} = 0.0016$$

$$\text{SegmentModulesWeight} = \text{ModuleWeight} \cdot 23 \text{ kg}$$

$$\text{SegmentModulesWeight} = 6.3940$$

$$\text{SegmentCasingHeight} = \text{EneSegHeight} \text{ m}$$

$$\text{EneSegmentDepth} = 6 \cdot \text{EneModDepth} \text{ m}$$

$$\text{EneSegmentWidth} = \text{EneModWidth} \cdot 4 \text{ m}$$

$$\text{garoliteThickness} = \frac{1}{16} \cdot \frac{25.4}{1000} \text{ m}$$

$$\text{SegmentModulesWeight} = \text{ModuleWeight} \cdot 23 \text{ kg}$$

Segment Cross Sectional Areas

$$\begin{aligned} \text{TopCrossSecArea} &= \text{EneSegmentDepth} \cdot \text{garoliteThickness} \cdot 2; \\ \text{WidthCrossSecArea} &= \text{EneSegmentDepth} \cdot \text{garoliteThickness} \cdot 2; \\ \text{DepthCrossSecArea} &= \text{EneSegmentWidth} \cdot \text{garoliteThickness} \cdot 2 + \\ &\quad \text{EneSegHeight} \cdot \text{garoliteThickness} \cdot 2; \\ \text{airHoleDia} &= 8 / 1000; \text{m} \end{aligned}$$

$$\text{TopCrossSecArea} = \text{EneSegmentDepth} \cdot \text{garoliteThickness} \cdot 2 \text{ m}^2$$

$$\text{WidthCrossSecArea} = \text{EneSegmentDepth} \cdot \text{garoliteThickness} \cdot 2 \text{ m}^2$$

$$\text{DepthCrossSecArea} = \text{EneSegmentWidth} \cdot \text{garoliteThickness} \cdot 2 + \text{EneSegHeight} \cdot \text{garoliteThickness} \cdot 2 \text{ m}^2$$

$$\text{airHoleDia} = \frac{8}{1000} \text{ m}$$

Tab Gluing Strength

$$\text{Pload} = 9.8 \cdot 20 \cdot \text{SegmentModulesWeight}$$

$$\text{Pload} = 1.2532 \text{e}+03$$

$$\text{insetlength} = 26.75 / 1000 \text{ m}$$

$$\text{insetlength} = 0.0267$$

$$\text{bolttosegmentlength} = 14 / 1000 \text{ m}$$

$$\text{bolttosegmentlength} = 0.0140$$

$$\text{L1} = 14 / 1000 \text{ m}$$

$$\text{L1} = 0.0140$$

$$L2 = 40.75/1000\text{m}$$

$$L2 = 0.0408$$

$$P1 = -(Pload*L2)/(L2-L1)$$

$$P1 = -1.9091\text{e}+03$$

$$P2 = (Pload*L1)/(L2-L1)$$

$$P2 = 655.8929$$

$$Mc = Pload*L2 - P1*(L2-L1)$$

$$Mc = 102.1378$$

$$Medge = -Pload*L1 + Mc$$

$$Medge = 84.5926$$

$$\text{Stress_bending} = (Medge*(garoliteThickness/2))/(4*garoliteThickness*garoliteThickness^3)/12$$

$$\text{Stress_bending} = 2.2025\text{e}+08$$

$$\begin{aligned} P_{\text{load}} &= 9.8 \times 20 \times \text{SegmentModulesWeight} \\ \text{insetlength} &= \frac{26.75}{1000} \text{ m} \\ \text{bolttosegmentlength} &= \frac{14}{1000} \text{ m} \\ L_1 &= \frac{14}{1000} \text{ m} \\ L_2 &= \frac{40.75}{1000} \text{ m} \\ P_1 &= \frac{-P_{\text{load}} \times L_2}{L_2 - L_1} \\ P_2 &= \frac{P_{\text{load}} \times L_1}{L_2 - L_1} \\ M_c &= P_{\text{load}} \times L_2 - P_1 \times (L_2 - L_1) \\ M_{\text{edge}} &= -P_{\text{load}} \times L_1 + M_c \\ \text{Stress_bending} &= \frac{M_{\text{edge}} \times \left(\frac{\text{garoliteThickness}}{2} \right)}{4 \times \text{garoliteThickness} \times \text{garoliteThickness}^3} \times \frac{1}{12} \end{aligned}$$

Lid (Polycarbonate) Properties

$$\text{polycarbonateThickness} = 1/4*25.4/1000$$

polycarbonateThickness = 0.0063

Lid Strength

Polycarbonate properties <https://www.matweb.com/search/DataSheet.aspx?MatGUID=501acbb63cbc4f748faa7490884cdbca>

$$M_a = (P_{load} \times E_{neSegmentDepth}^2) / 12$$

$$M_a = 18.1602$$

$$\text{Stress_bending} = (M_a \times (\text{polycarbonateThickness} / 2)) / (\text{polycarbonateThickness}^3 \times (E_{neSegmentWidth}) / 12)$$

$$\text{Stress_bending} = 3.3279e+07$$

$$\begin{aligned} \text{PolycarbonateStrength} &= 62000000; \\ S_{fLid} &= \text{PolycarbonateStrength} / \text{Stress_bending} \end{aligned}$$

$$S_{fLid} = 1.8630$$

$$\text{PolycarbonateFlexuralModulus} = 2228000000$$

$$\text{PolycarbonateFlexuralModulus} = 2.2280e+09$$

$$M_a = \frac{P_{load} \times E_{neSegmentDepth}^2}{12}$$

$$\text{Stress_bending} = \frac{M_a \times \left(\frac{\text{polycarbonateThickness}}{2} \right)}{\text{polycarbonateThickness}^3 \times E_{neSegmentWidth}} \times \frac{1}{12}$$

$$\text{PolycarbonateStrength} = 62000000$$

$$S_{fLid} = \frac{\text{PolycarbonateStrength}}{\text{Stress_bending}}$$

Lid Deflection from 20g Loading

$$\begin{aligned} \text{Packload} &= \text{TopCrossSecForce} ; \\ E_{neLidSegmentDepth} &= 6 \times E_{neModDepth} + 0.082 - 0.022; \% \\ v_{max} &= \text{Packload} * (E_{neLidSegmentDepth}^4) / 384 / (\text{PolycarbonateFlexuralModulus}) / \\ &(((E_{neSegmentWidth}) * \text{polycarbonateThickness}^3) / 12) \end{aligned}$$

$$v_{max} = 0.0438$$

$$\text{Packload} = \text{TopCrossSecForce} \times \text{N}$$

$$\text{EneLidSegmentDepth} = 6 \times \text{EneModDepth} + 0.082 - 0.022 \text{ m}$$

$$v_{\max} = \frac{\text{Packload} \times \text{EneLidSegmentDepth}^4}{384 \times \text{garoliteFlexuralModulus} \times \left(\frac{\text{EneSegmentWidth} \times \text{garoliteThickness}^3}{12} \right)}$$

Garolite Properties

```
psiTopPascal = 6894.76;
garoliteStrength = 38000*psiTopPascal;%262,000,000pa
garoliteFlexuralModulus = 2400* 1000 * psiTopPascal
```

```
garoliteFlexuralModulus = 1.6547e+10
```

Segment Forces and Stresses

```
TopCrossSecForce = SegmentModulesWeight*20*9.8%newtons
```

```
TopCrossSecForce = 1.2532e+03
```

```
WidthCrossSecForce = 6*SegmentModulesWeight*40*9.8;%newtons
DepthCrossSecForce = SegmentModulesWeight * 40 * 9.8;%newtons
```

```
TopStress = TopCrossSecForce/TopCrossSecArea%pa
```

```
TopStress = 9.4656e+05
```

```
WidthStress = WidthCrossSecForce/WidthCrossSecArea%pa
```

```
WidthStress = 1.1359e+07
```

```
DepthStress = DepthCrossSecForce/DepthCrossSecArea%pa
```

```
DepthStress = 4.2261e+06
```

```
SafetyFactorCasing = garoliteStrength/WidthStress%pa
```

```
SafetyFactorCasing = 23.0660
```

$$\text{TopCrossSecForce} = \text{SegmentModulesWeight} \times 20 \times 9.8 \text{ N}$$

$$\text{WidthCrossSecForce} = 6 \times \text{SegmentModulesWeight} \times 40 \times 9.8 \text{ N}$$

$$\text{DepthCrossSecForce} = \text{SegmentModulesWeight} \times 40 \times 9.8 \text{ N}$$

$$\text{TopStress} = \frac{\text{TopCrossSecForce}}{\text{TopCrossSecArea}} \text{ Pa}$$

$$\text{WidthStress} = \frac{\text{WidthCrossSecForce}}{\text{WidthCrossSecArea}} \text{ Pa}$$

$$\text{DepthStress} = \frac{\text{DepthCrossSecForce}}{\text{DepthCrossSecArea}} \text{ Pa}$$

$$\text{SafetyFactorCasing} = \frac{\text{garoliteStrength}}{\text{WidthStress}} \text{ Pa}$$

Vertical Minimum Column Properties

```
vmincolumnlength = EneSegmentWidth*1000%mm
```

```
vmincolumnlength = 81.2000
```

```
vmincolumnarea = (garoliteThickness*EneSegHeight -  
5*airHoleDia*garoliteThickness)*1000^2%mm^2
```

```
vmincolumnarea = 104.1400
```

```
vmincolumnMOI = ((garoliteThickness^3*(EneSegmentWidth-5*airHoleDia))/  
12)*1000^4%mm^4
```

```
vmincolumnMOI = 13.7359
```

```
vPcrEuler = (pi^2*garoliteFlexuralModulus*vmincolumnMOI)/vmincolumnlength^2
```

```
vPcrEuler = 3.4023e+08
```

```
stress_compressive = TopCrossSecForce/(vmincolumnarea/1000^2)
```

```
stress_compressive = 1.2034e+07
```

$$vmincolumnlength = EneSegmentWidth \times 1000 \text{ mm}$$

$$vmincolumnarea = (garoliteThickness \times EneSegHeight - 5 \times airHoleDia \times garoliteThickness)$$

$$vmincolumnMOI = \left(\frac{garoliteThickness^3 \times (EneSegmentWidth - 5 \times airHoleDia)}{12} \right) \times 1000$$

$$vPcr_{Euler} = \frac{\pi^2 \times garoliteFlexuralModulus \times vmincolumnMOI}{vmincolumnlength^2}$$

$$stress_compressive = \frac{TopCrossSecForce}{\left(\frac{vmincolumnarea}{1000^2} \right)}$$

Lateral Minimum Column Properties

$$lmincolumnlength = EneSegmentDepth \times 1000 \text{ mm}$$

$$lmincolumnlength = 417.0000$$

$$lmincolumnarea = (garoliteThickness \times (EneSegmentWidth + 2 \times garoliteThickness)) \times 1000^2 \text{ mm}^2$$

$$lmincolumnarea = 133.9453$$

$$lmincolumnMOI = \left(\frac{garoliteThickness^3 \times (EneSegmentWidth + 2 \times garoliteThickness)}{12} \right) \times 1000^4 \text{ mm}^4$$

$$lmincolumnMOI = 28.1303$$

$$lmincolumnlength = EneSegmentDepth \times 1000 \text{ mm}$$

$$lmincolumnarea = (garoliteThickness \times (EneSegmentWidth + 2 \times garoliteThickness)) \times 1000^2$$

$$lmincolumnMOI = \left(\frac{garoliteThickness^3 \times (EneSegmentWidth + 2 \times garoliteThickness)}{12} \right) \times 1000^4$$

Bolt calculations

Tensile Stress Area

If a threaded rod as shown in Figure 15-2 is subjected to pure tensile loading, one might expect its strength to be limited by the area of its minor (root) diameter, d_r . However, testing of threaded rods in tension shows that their tensile strength is better defined by the average of the minor and pitch diameters. A *tensile-stress area* A_t is defined as

$$A_t = \frac{\pi}{4} \left(\frac{d_p + d_r}{2} \right)^2 \quad (15.1a)$$

where, for UNS threads:

$$d_p = d - 0.649519/N \quad d_r = d - 1.299038/N \quad (15.1b)$$

and for ISO threads:

$$d_p = d - 0.649519p \quad d_r = d - 1.226869p \quad (15.1c)$$

with d = outside diameter, N = number of threads per inch, and p = pitch in mm.

The stress in a threaded rod due to a pure axial tensile load F is then

$$\sigma_t = \frac{F}{A_t} \quad (15.2)$$

Table 15-2 Principal Dimensions of ISO Metric Threads
Data Calculated from Equations 15.1—

Coarse Threads			
Major Diameter d (mm)	Pitch p (mm)	Minor Diameter d_r (mm)	Tensile Stress Area A_t (mm ²)
3.0	0.50	2.39	5.03
3.5	0.60	2.76	6.78
4.0	0.70	3.14	8.78
5.0	0.80	4.02	14.18
6.0	1.00	4.77	20.12
7.0	1.00	5.77	28.86

```
dia = 6;
pitch = 1;
diaPitch = dia - 0.649519*pitch;%mm
diaRoot = dia - 1.226869*pitch;%mm
AreaBoltTensile = pi/4*((diaPitch+diaRoot)/2)^2;%in mm^2
ABTMeters = AreaBoltTensile/1000000;%Converts to Meters
exampleGr5Tensile = 827370875;%Converted from psi
MaxTensileForFastener = exampleGr5Tensile*ABTMeters %newtons
```

```
MaxTensileForFastener = 1.6649e+04
```

```
SafetyFactorBoltTension = MaxTensileForFastener*2/TopCrossSecForce%assumes all load
is on one side of segment
```

```
SafetyFactorBoltTension = 26.5707
```

```
SafetyFactorVultShear = MaxTensileForFastener*2/WidthCrossSecForce%Assumes
```

```
SafetyFactorVultShear = 2.2142
```

```
% no preload friction an only one side
```

$$\text{dia} = 6 \text{ mm}$$

$$\text{pitch} = 1 \text{ mm}$$

$$\text{diaPitch} = \text{dia} - 0.649519 \times \text{pitch} \text{ mm}$$

$$\text{diaRoot} = \text{dia} - 1.226869 \times \text{pitch} \text{ mm}$$

$$\text{AreaBoltTensile} = \frac{\pi}{4} \left(\frac{\text{diaPitch} + \text{diaRoot}}{2} \right)^2 \text{ mm}^2$$

$$\text{ABTMeters} = \frac{\text{AreaBoltTensile}}{1000000} \text{ m}^2$$

$$\text{exampleGr5Tensile} = 827370875 \text{ Pa}$$

$$\text{MaxTensileForFastener} = \text{exampleGr5Tensile} \times \text{ABTMeters} \text{ N}$$

$$\text{SafetyFactorBoltTension} = \frac{\text{MaxTensileForFastener} \times 2}{\text{TopCrossSecForce}}$$

$$\text{SafetyFactorBoltShear} = \frac{\text{WidthCrossSecForce} \times 12}{\text{MaxTensileForFastener}}$$

Tear out

<https://laminatedplastics.com/g-10.pdf>

```
BoltDistanceFromEdge = 8/1000;%mm realistic distance from edge
maxTearoutForce = BoltDistanceFromEdge*garoliteThickness*garoliteStrength
```

```
maxTearoutForce = 3.3274e+03
```

$$\text{BoltDistanceFromEdge} = \frac{8}{1000} \text{ m}$$

$$\text{maxTearoutForce} = \text{BoltDistanceFromEdge} \cdot \text{garoliteThickness} \cdot \text{garoliteStrength} \text{ N}$$

Flange Bonding(Pure Pull)

```
fr4BondStrength = 2200*psiTopPascal;
MinFlangeBondArea = DepthCrossSecForce/fr4BondStrength
```

```
MinFlangeBondArea = 1.6524e-04
```

```
DepthRestrictingBondArea = garoliteThickness * 80/1000
```

```
DepthRestrictingBondArea = 1.2700e-04
```

```
SideBondArea = 0.02* garoliteThickness*2
```

```
SideBondArea = 6.3500e-05
```

$$\text{bondMaxForce} = \text{fr4BondStrength} * (\text{DepthRestrictingBondArea} + \text{SideBondArea})$$

$$\text{bondMaxForce} = 2.8896\text{e}+03$$

$$\text{fr4BondStrength} = 2200 \times \text{psiTopPascal Pa}$$

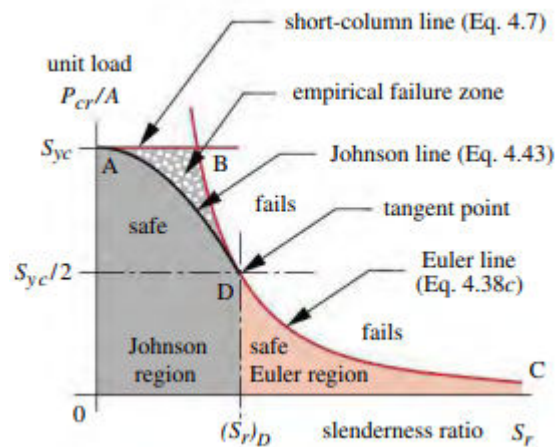
$$\text{MinFlangeBondArea} = \frac{\text{DepthCrossSecForce}}{\text{fr4BondStrength}} \text{ m}^2$$

$$\text{DepthRestrictingBondArea} = \text{garoliteThickness} \times \frac{80}{1000} \text{ m}^2$$

$$\text{SideBondArea} = 0.02 \times \text{garoliteThickness} \times 2 \text{ m}^2$$

$$\text{bondMaxForce} = \text{fr4BondStrength} \times (\text{DepthRestrictingBondArea} + \text{SideBondArea})$$

Euler-Johnson buckling limits



$$I = \sum_{i=0}^n (I)_i = \sum_{i=0}^n (\bar{I} + Ad^2)_i$$

$$\begin{aligned} \text{MomentOfInertiaSideWalls} &= 2 * ((\text{EneSegmentWidth}/2 + \text{garoliteThickness}/2)^2 \\ &* \text{garoliteThickness} * (\text{EneSegHeight} + (2 * \text{garoliteThickness})) + \\ &(\text{garoliteThickness}^3 * \text{SegmentCasingHeight}/12) \end{aligned}$$

$$\text{MomentOfInertiaSideWalls} = 5.9179\text{e}-07$$

% half width to get to walls + half of garolite thickness = centroid
 % distance away, * area sidewall + garolite thickness^3(height) * segment
 height(width)

$$\begin{aligned} \text{MomentOfInertiaBottom} &= (\text{garoliteThickness}/2 + \text{EneSegHeight}/2)^2 * \\ &\text{garoliteThickness} * \text{EneSegmentWidth} + (\text{garoliteThickness}^3 * \text{EneSegmentWidth}/12) \end{aligned}$$

$$\text{MomentOfInertiaBottom} = 3.7028\text{e}-07$$

$$\begin{aligned} \text{MomentOfInertiaLid} &= (\text{garoliteThickness}/2 + \text{EneSegHeight}/2 + (15/1000))^2 * \\ &\text{garoliteThickness} * \text{EneSegmentWidth} + (\text{garoliteThickness}^3 * \text{EneSegmentWidth}/12) \end{aligned}$$

MomentOfInertiaLid = 6.0654e-07

$$i_{xx} = \text{MomentOfInertiaLid} + \text{MomentOfInertiaBottom} + \text{MomentOfInertiaSideWalls}$$

$i_{xx} = 1.5686e-06$

$$\text{RadiusOfGyration} = \sqrt{i_{xx} / \text{DepthCrossSecArea}}$$

RadiusOfGyration = 0.0514

$$\text{SlendernessRatioCasing} = \text{EneSegmentDepth} / \text{RadiusOfGyration}$$

SlendernessRatioCasing = 8.1085

$$\text{SlendernessRatioForHalfStrength} = \pi \cdot \sqrt{2 \cdot \text{garoliteFlexuralModulus} / \text{garoliteStrength}}$$

SlendernessRatioForHalfStrength = 35.3085

$$\text{MaxForceJohnson} = \text{DepthCrossSecArea} \cdot (\text{garoliteStrength} - (1 / \text{garoliteFlexuralModulus} \cdot (\text{garoliteStrength} \cdot \text{SlendernessRatioCasing} / (2 \cdot \pi))^2))$$

MaxForceJohnson = 1.5129e+05

$$\text{SFJohnson} = \text{MaxForceJohnson} / \text{WidthCrossSecForce}$$

SFJohnson = 10.0602

$$\text{MomentOfInertiaSideWalls} = 2 \left(\left(\frac{\text{EneSegmentWidth}}{2} + \frac{\text{garoliteThickness}}{2} \right)^2 \cdot \text{garoliteThickness} \cdot (\text{EneSegHeight} + \frac{\text{garoliteThickness}^3 \cdot \text{SegmentCasingHeight}}{12}) \right) \text{ m}^4$$

$$\text{MomentOfInertiaBottom} = \left(\frac{\text{garoliteThickness}}{2} + \frac{\text{EneSegHeight}}{2} \right)^2 \cdot \text{garoliteThickness} \cdot \text{EneSegmentWidth} + \frac{\text{garoliteThickness}^3 \cdot \text{EneSegmentWidth}}{12} \text{ m}^4$$

$$\text{MomentOfInertiaLid} = \left(\frac{\text{garoliteThickness}}{2} + \frac{\text{EneSegHeight}}{2} + 0.015 \right)^2 \cdot \text{garoliteThickness} \cdot \text{EneSegmentWidth} + \frac{\text{garoliteThickness}^3 \cdot \text{EneSegmentWidth}}{12} \text{ m}^4$$

$$I_{xx} = \text{MomentOfInertiaLid} + \text{MomentOfInertiaBottom} + \text{MomentOfInertiaSideWalls} \text{ m}^4$$

$$\text{RadiusOfGyration} = \sqrt{\frac{I_{xx}}{\text{DepthCrossSecArea}}} \text{ m}$$

$$\text{SlendernessRatioCasing} = \frac{\text{EneSegmentDepth}}{\text{RadiusOfGyration}}$$

$$\text{SlendernessRatioForHalfStrength} = \pi \cdot \sqrt{\frac{2 \cdot \text{garoliteFlexuralModulus}}{\text{garoliteStrength}}}$$

$$\text{MaxForceJohnson} = \text{DepthCrossSecArea} \cdot \left(\text{garoliteStrength} - \frac{1}{\text{garoliteFlexuralModulus}} \cdot \left(\frac{\text{garoliteStrength}^3}{\pi^2} \right) \right)$$

$$\text{SFJohnson} = \frac{\text{MaxForceJohnson}}{\text{WidthCrossSecForce}}$$

Passive Thermal Properties

```
SurfaceArea = EneSegmentDepth*EneSegmentWidth*2 + EneSegHeight*EneSegmentDepth*2;
WattsPerKelvin = 1/garoliteThickness * 0.288 * SurfaceArea % W/m/K * m^2 /m = w/k
```

WattsPerKelvin = 28.2632